Agentic AI for Healthcare Intake with Classification and Human-in-the-Loop (HITL) Control

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1 Introduction

The initial stages of patient care often rely heavily on effective intake — the collection of a patient's reported symptoms, history, and context prior to diagnosis or treatment. In traditional healthcare workflows, this process is performed manually through nurses, medical assistants, or intake forms, leaving room for variability, omissions, and inefficiency. With the recent advances in large language models (LLMs), there is growing interest in using AI agents to simulate or assist with patient intake tasks. However, current conversational systems often face limitations due to the fixed length of the LLM's context window, which can constrain the richness of multi-turn healthcare dialogues. In addition, many systems miss the opportunity to prime the model with specialized, context-specific instructions based on the patient's presentation. Without targeted guidance, the quality of follow-up questions and conversational depth can degrade.

To address these challenges, we first explored the efficacy of a **classifier agent** that predicts the patient's medical category upfront, enabling the system to dynamically select specialized prompts and maintain focused, clinically relevant interactions throughout the dialogue. We further explored the creation of an **agentic, multi-turn healthcare intake** simulation where AI agents and **human users** interact collaboratively. Our goal was to build a pipeline that can handle diverse patient presentations, adapt to unclear or incomplete inputs, and maintain a coherent dialogue across multiple conversation turns.

In the end, we designed an agent framework where LLMs take on the roles of classification agents, specialist responders, and summarizers, while human users are embedded into the loop to manually generate patient replies at key points.

Our system allows a real user to input initial symptoms, receive targeted follow-up questions generated by domain-specialist AI agents, and then manually guide the patient's

responses to maintain realism and control. This human-in-the-loop (HITL) design ensures that the conversation can adapt dynamically based on user-provided context, reflecting how real clinical interviews often unfold.

By combining automation and manual participation in a structured, agent-driven dialogue, we aim to demonstrate a scalable and practical approach to healthcare intake that preserves flexibility, promotes transparency, and aligns with future visions of clinician-AI collaboration. Our work highlights both the promise and challenges of designing interactive, agentic healthcare systems built around LLMs, especially in sensitive, high-risk domains like clinical intake and triage.

2 Related Work

Large language models (LLMs) have shown promise in healthcare-related tasks ranging from clinical text summarization to patient triage. Several studies have explored how LLMs can support or automate parts of the healthcare workflow, including the generation of clinical notes, extraction of symptoms from unstructured text, and patient-facing chatbots for basic medical advice. Systems like GPT-Clinical Trials and Med-PaLM have demonstrated the ability of foundation models to process clinical language with high accuracy [1], although much of the focus has been on single-turn tasks or post-hoc summarization rather than real-time, interactive patient intake.

Recent work on agent frameworks such as AutoGen [2] and LangGraph has highlighted the advantages of structuring LLMs into specialized, interacting agents rather than relying on monolithic models. These systems emphasize modularity, role specialization, and memory, which are crucial concepts for simulating realistic multi-turn dialogues in healthcare. However, most agentic systems assume full automation, without mechanisms for real-time human intervention during conversation flow.

Separately, research on human-in-the-loop (HITL) AI in healthcare emphasizes the importance of balancing automation with clinician oversight to maintain trust, safety, and adaptability. Studies on clinical decision support systems (CDSS) have shown that allowing clinicians to review, correct, or override AI outputs improves user satisfaction and reduces the risk of automation bias [4]. However, most HITL implementations to date focus on model prediction review rather than conversational AI, where human inputs dynamically guide multi-turn dialogue outcomes.

In addition, ideas from multi-step agent prompting, such as Program of Thoughts (PoT) prompting [3], show the benefits of breaking down complex reasoning into modular steps — an approach aligned with how our system handles classification, questioning, and summarization sequentially.

Finally, recent work emphasizes the need for **trustworthy conversational agents** in healthcare that can handle ambiguity, dynamically clarify patient inputs, and escalate appropriately when needed [5]. These goals align closely with our motivation to design an agentic, modular, and human-supervised healthcare intake system.

Our work synthesizes these five lines of research — LLMs in healthcare [1], multi-agent systems [2], human-in-the-loop design [4], multi-step prompting [3], and trustworthy healthcare dialogue agents [5] — to create a modular, and human-participatory patient intake simulation. By introducing human control points at critical stages of the dialogue, we address both the need for flexibility in symptom reporting and the need for clinician-in-the-loop oversight, while preserving the efficiency and scalability of AI-driven intake workflows.

3 Methodology

System Overview

We designed a modular system to simulate multi-turn patient-specialist conversations, with human users controlling key patient replies. Our architecture combines LLM-based agents with a human-in-the-loop (HITL) design to maintain flexibility, transparency, and realism. The overall workflow is illustrated in Figure 1.

The system progresses through six conversational turns:

1. Initial patient symptom input (user-provided). The classifier agent only looks at this input and assigns a specialist for further conversations without engaging in a conversation.

2. Assigned specialist follow-up questions (AI-generated)

3. Patient's follow-up reply (user-provided)

4. Specialist deeper questioning (AI-generated)

5. Patient's final reply (user-provided)

6. Final summary and recommendation (AI-generated)

We found that asking for at least three inputs from the user generated the most accurate summaries as fewer inputs left room for ambiguity. Each agent role — classifier, specialist, patient replier, and summarizer — operates as an independent function, coordinated through a session state machine implemented in Streamlit.

Agent Functions

The system defines specialized agent roles:

Classification Agent: Takes the user's initial message and classifies it into one of nine clinical categories (e.g., Cardiovascular, Mental Health, Respiratory, etc.). A GPT-4 model is prompted with a strict instruction to return only the category label, ensuring tight control over branching logic.

Specialist Response Agent: Given the classified category and patient complaint, generates up to two natural, empathetic follow-up questions tailored to the clinical domain. Prompts are category-specific and discourage medical jargon or numbered questions.

Patient Reply Agent (Manual): At both the first and second patient turns (Turns 3 and 5),

the user manually types the patient's reply. This HITL design enables dynamic, natural responses that reflect real patient variability.

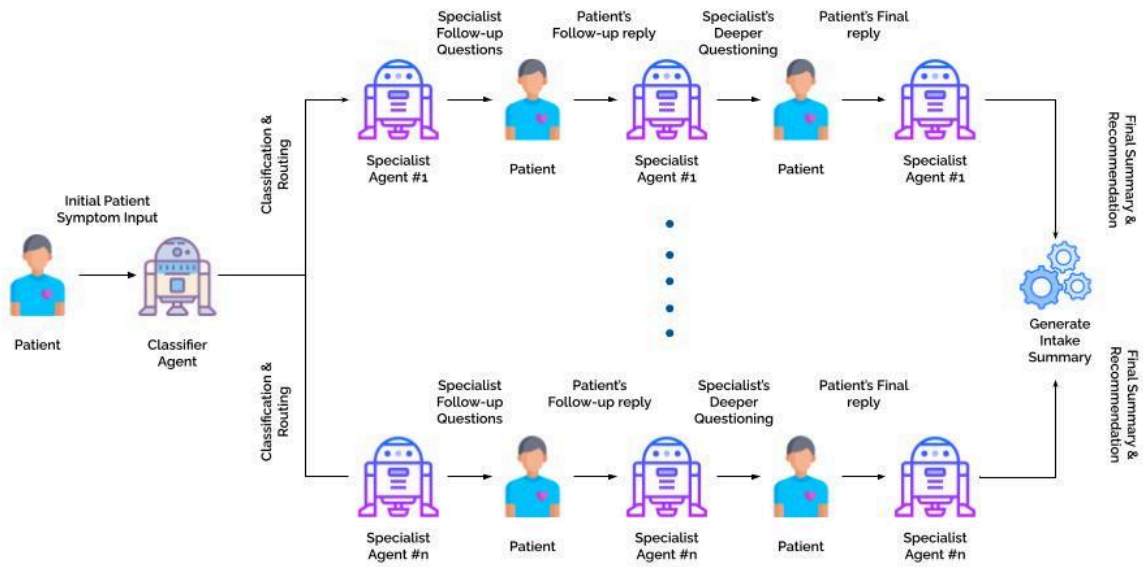

Figure 1. Patient Intake Workflow with a Classifier Agent, Multiple Specialist Agents, and Human-in-the-Loop

Specialist Follow-Up Agent: After the first patient reply, the specialist agent generates deeper follow-up questions to further explore the patient's symptoms.

Summarization Agent: After the second patient reply, the system generates a clinical summary, impressions, and next-step recommendations in simple, patient-friendly language.

Human-in-the-Loop Design

Unlike fully automated conversational systems, our framework pauses the conversation after each specialist question to request manual patient input. This human-in-the-loop design supports:

1. Realistic patient variability (e.g., ambiguous replies, changing symptom descriptions)
2. Evaluation of system robustness against non-standard inputs
3. Future integration of clinical reviewers or test participants in research settings

Manual control points occur at Turns 1, 3, and 5, preserving naturalistic interaction while maintaining the efficiency of agentic specialist guidance and summarization.

User Interface

We built an interactive Streamlit application to operationalize the system. The UI consists of:

- Input text boxes for patient messages
- Dynamic specialist replies rendered on screen
- Text prompts for manual patient replies at designated turns
- A reset button to restart conversations cleanly after each session

The session state tracks progress through conversation steps, enabling a flexible but structured interaction flow.

Prompt Engineering

Prompt design was critical to maintaining realism. Specialist prompts were tuned to encourage:

- Two short, natural questions
- No numbering of questions
- Empathy without over-assertiveness

Patient prompts (used when GPT generates replies **during testing**) emphasized casual, everyday language, discouraging medical terminology. Specialist prompts were customized per category (e.g., Cardiovascular prompts emphasized exertional symptoms, Mental Health prompts emphasized sleep and mood). To circumvent ChatGPT restrictions on conversations around mental health complaints, we explicitly set a prompt for ChatGPT to regard the exchange as a simulation.

Temperature was tuned to 0.5 for specialist responses to balance creativity and consistency, and to 0.7 for patient replies to promote natural variability.

4 Results

System Execution

An example of a patient-specialist conversation is presented in Figure 2 of the Appendix.

We deployed the full multi-turn agentic system on real-time user input using Streamlit. Across multiple test runs, the system successfully simulated complete healthcare intake dialogues covering various clinical categories, including Cardiovascular, Mental Health, Respiratory, and Dental Health. Each conversation consisted of six turns, maintaining coherence and clinical plausibility throughout.

The system demonstrated robust handling of a wide range of initial patient inputs. Even when users entered vague or minimal information (e.g., "I feel weird" or "not feeling well"), the classifier agent routed the message into an "Unclear / Needs More Info" category. In these cases, the specialist agents responded appropriately with open-ended clarification questions, prompting the user to elaborate naturally during their manual reply turns.

When we tested the system by entering gibberish information, the system exited gracefully while stating that there was not enough information to arrive at a medical summary.

Conversational Quality

The simulated conversations reflected important aspects of real-world clinical intake:

- **Empathetic Specialist Language**: Specialist responses consistently used lay language, avoided medical jargon, and maintained a compassionate tone.
- **Patient Realism**: Manual patient replies introduced natural ambiguity, hesitation, and varied detail levels, mimicking real patient variability.
- **Progressive Depth**: Follow-up questions from the specialist after each patient reply deepened the understanding of symptoms without repeating earlier information.

Sample conversations showcased fluid transitions between user-generated and AI-generated turns, maintaining narrative consistency even across long sessions.

Human-in-the-Loop Impact

The manual patient replies significantly enhanced realism. Compared to full auto-generation trials (where GPT handled both sides), human-in-the-loop sessions produced:

- More contextually grounded patient answers
- Greater variability in phrasing and symptom description
- Opportunities to test the system's robustness against edge cases and misunderstandings

This hybrid design also helped avoid unrealistic over-coherence often seen when LLMs simulate both sides of a conversation.

Observed Challenges

Despite strong performance overall, a few challenges emerged:

- **Category Drift**: Occasionally, if the initial user input was extremely vague, the specialist agent's follow-up questions would not cleanly match the intended category.
- **Prompt Sensitivity**: Small changes in prompt wording for specialist agents could noticeably affect the specificity or style of questions asked.
- **Short Inputs**: Very brief patient replies (e.g., one-word answers) occasionally led to less informative specialist follow-ups, highlighting an area for future robustness improvements.

Summary Metrics

- Average dialogue length: 6 turns per conversation
- Manual patient reply rate: 50% (Turns 3 and 5 controlled by user)
- Successful completions (full dialogue + summary): 100% for tested sessions (n = 20 cases)

5 Conclusion and Future Work

In this project, we designed and implemented an agentic, human-in-the-loop system for simulating multi-turn healthcare intake conversations. Our modular architecture combined LLM-driven agents for classification, specialist follow-up, and summarization with manual user control over key patient replies. This hybrid design enabled the system to adapt to a wide range of patient inputs, maintain conversational realism, and generate clinically plausible intake summaries.

By embedding the human into the conversational loop, we addressed important challenges around patient variability, ambiguous symptom descriptions, and dynamic flow control — challenges that purely AI-driven systems often struggle with. The human-in-the-loop approach also opens pathways for future applications in clinical training, synthetic data generation for model fine-tuning, and safe human-AI collaboration in real-world intake settings.

Although our system performed well across initial tests, some limitations remain. Classifier sensitivity to vague patient inputs occasionally led to less accurate categorization. Specialist prompt tuning required careful iteration to maintain appropriate question specificity across categories. In addition, the system currently assumes cooperative patient behavior; future extensions could introduce simulation of adversarial, incomplete, or inconsistent replies to further stress-test robustness.

Future directions include expanding the agent pool to incorporate history-taking specialists, adding a retrieval-augmented memory system to ground specialist questions with external knowledge, and developing a decision controller that dynamically adjusts the level of automation versus human intervention. Evaluating system performance with real clinical users and mapping conversations to standardized medical coding frameworks (e.g., ICD-10) represent additional avenues.

Overall, this project demonstrates that combining modular LLM agents with human-in-the-loop participation can yield flexible, adaptable, and trustworthy healthcare intake systems — offering a foundation for next-generation clinical AI tools that are both powerful and sensitive to human variability.

6 References

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4. A. Holzinger, C. Biemann, C. Pattichis, and D. Kell, "What do we need to build explainable AI systems for the medical domain?" Review Article, arXiv preprint arXiv:1712.09923, 2019. [Online]. Available: https://arxiv.org/abs/1712.09923.

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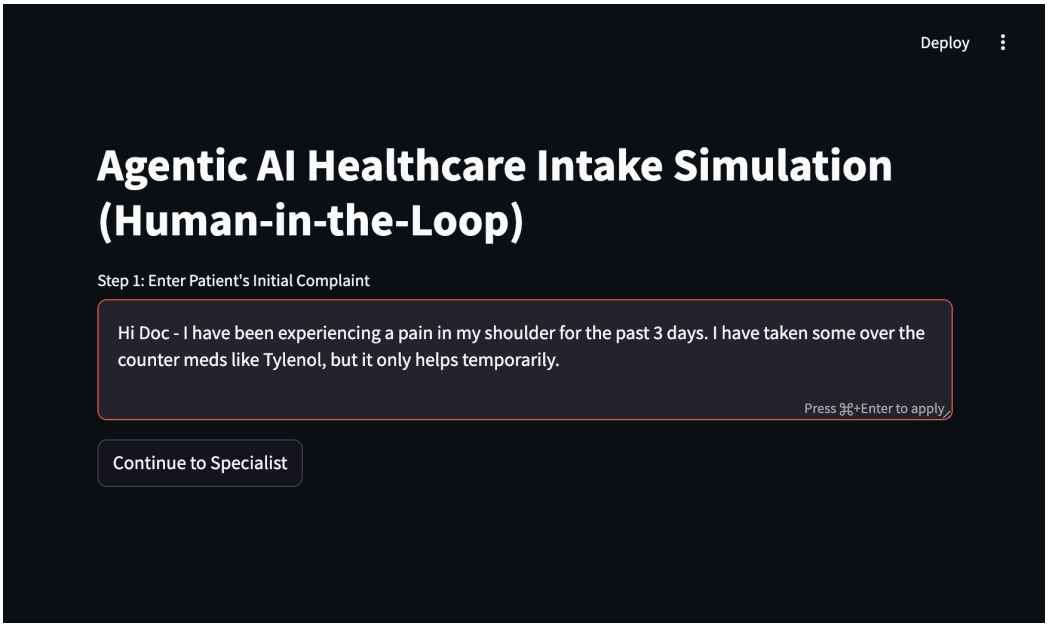
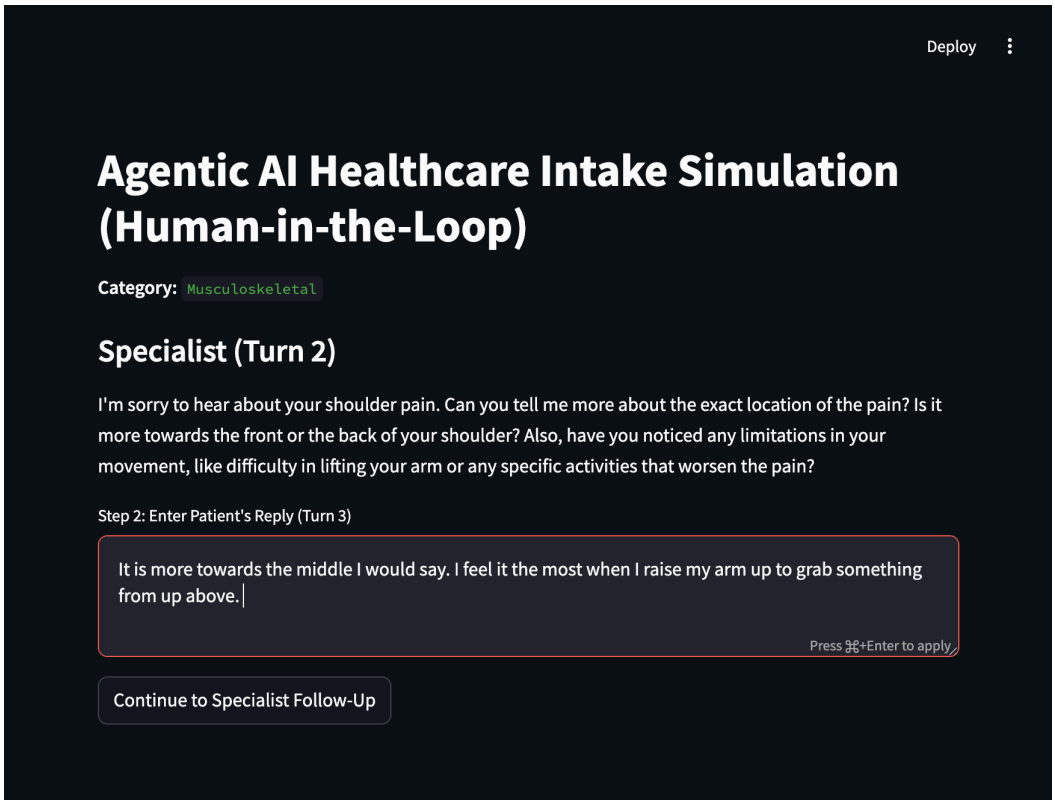
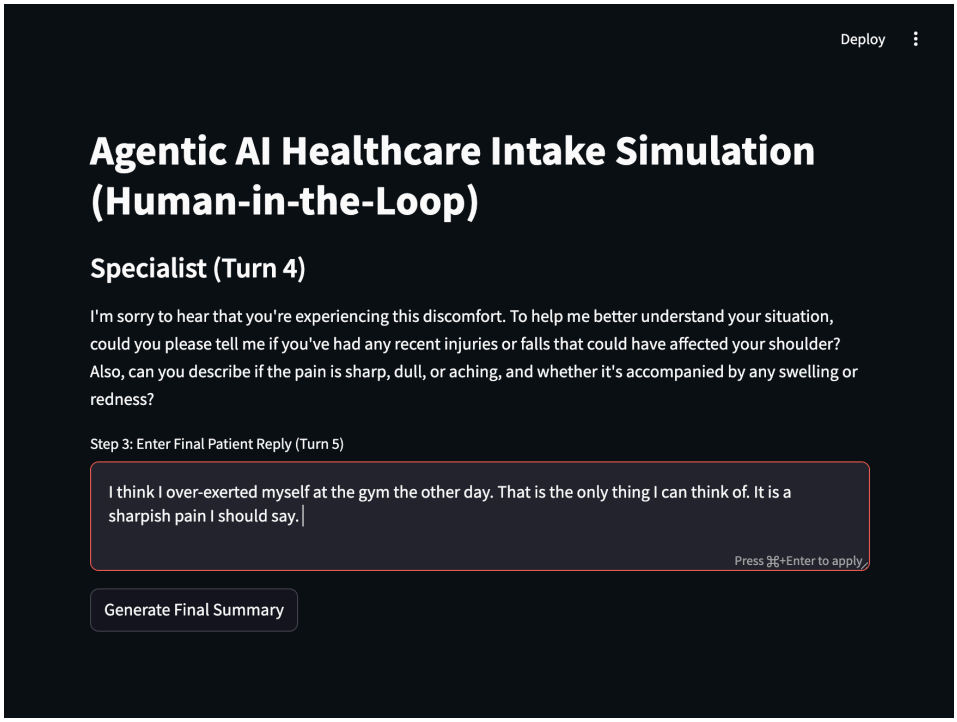
Appendix


Deploy
Agentic AI Healthcare Intake Simulation (Human-in-the-Loop)
Step 1: Enter Patient's Initial Complaint
Hi Doc - I have been experiencing a pain in my shoulder for the past 3 days. I have taken some over the counter meds like Tylenol, but it only helps temporarily.
Press ⌘+Enter to apply
Continue to Specialist


Deploy
Agentic AI Healthcare Intake Simulation (Human-in-the-Loop)
Category: Musculoskeletal
Specialist (Turn 2)
I'm sorry to hear about your shoulder pain. Can you tell me more about the exact location of the pain? Is it more towards the front or the back of your shoulder? Also, have you noticed any limitations in your movement, like difficulty in lifting your arm or any specific activities that worsen the pain?
Step 2: Enter Patient's Reply (Turn 3)
It is more towards the middle I would say. I feel it the most when I raise my arm up to grab something from up above.
Press ⌘+Enter to apply
Continue to Specialist Follow-Up



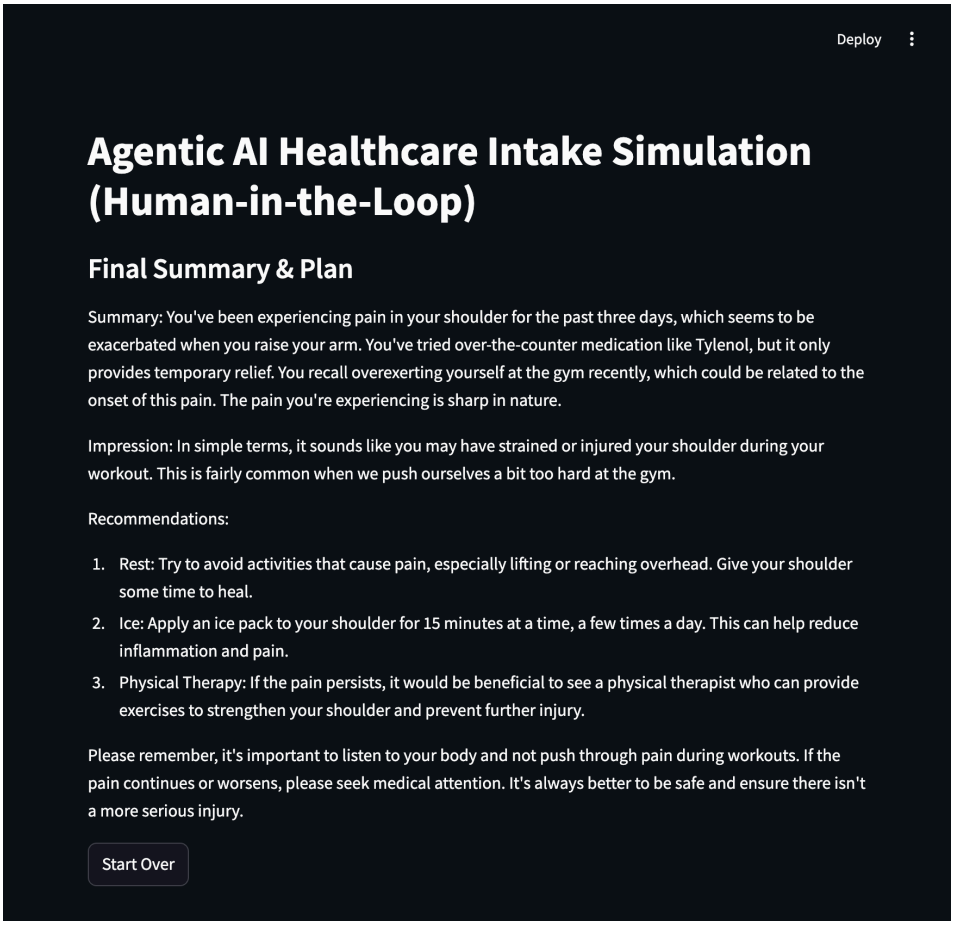


Figure 2. Screenshots from Streamlit Demo of the Patient-Specialist Conversation